

ECE 310 – Fall 2025

Lab 8

Summary

In this lab, you will implement a sequential 8×8 unsigned multiplier using behavioral modeling.

Deliverables

1. Pen-and-paper design of multiplier.
2. Code for model.
3. Screenshots of simulation waveforms with at least 2 sets of non-trivial 8-bit numbers (i.e. at least 2 multiplication results are needed).
4. Screenshots of synthesis results (*RTL Analysis* entry in Vivado) and discussion comparing to expected results in Step 1.
5. Extra Credit – up to 10 additional points will be added for typesetting your report in LaTeX, depending on usage of LaTeX elements (e.g. figures, tables, sections, etc).

Instructions

1. A sequential multiplier is a multiplier circuit that uses the shift-and-add method to produce a result, as follows:
 - a. If the multiplier LSB is '1', add the multiplicand value to the current running product value. If LSB is '0', do nothing.
 - b. Shift product and multiplicand registers left by 1 bit and multiplier right by 1 bit.
 - c. Repeat the above steps n times.
2. Create a project named **mult_seq_8x8** and a module with the same name with the following port connections:
 - a. Inputs: **multiplier**, **multiplicand** – 8 bits
 - b. Outputs: **product** – 16 bits
 - c. Other signals: **reset** (1 bit, active-high synchronous), **clock** (1 bit)
3. Use behavioral modeling in implementing your design. Additional internal signals may be used inside your module implementation. You do not need to show the design diagram at the gate-level, but the general operation and the units used for implementation should be clearly understood from your diagram. The reset signal will be asserted any time there is a new input to be processed and will initialize the product output to 0's.

Grading breakdown:

- Pen-and-paper/digital designs and comparison – 20 pts
- Sequential multiplier model/implementation in Verilog – 40 pts
- Passing tests – 20 pts
- Lab report with as much information present as possible, including test vectors and reasoning, resulting waveforms and discussion, synthesis vs. paper design results and discussion – 20 pts
- Extra Credit – up to 10 additional points will be added for typesetting your report in LaTeX, depending on usage of LaTeX elements (e.g. figures, tables, sections, etc).